

Wire Repair Instructions:

- Determine the level of damage to the wire
- Depending on damage and circumstance, it may be possible to redo a mechanical connection or there may have to be another electrical connection made
- A mechanical connection consists of a good mechanical joint between wires such that it can withstand moderate levels of wear
- An electrical connection consists of solder or connectors which are used to join the wires together solder is preferred for a more permanent solution while connectors are used for a connection where disconnecting wires might be necessary

PROCEDURE:

To Solder Wires Together:

- In case the same wire length is to be used, clean the wires of rust and other impurities before soldering
- Make a solid mechanical connections by splicing the wires together and twisting them (other methods can also be employed)
- Let the soldering iron heat up and tin the tip using standard 60-40 rosin core solder
- Start heating the joint and slowly add solder to the **JOINT** and not the iron, this helps make a solid connection as the solder seeps into the junction making it permanent
- Let the solder flow over the entire joint and let it cool
- Add heat shrink or insulation tape to insulate and protect the joint from the environment

To Crimp Connectors To Wires:

- There are two types of connectors that are usually used in this case:
Ferrules (screw terminals, wago, phoenix block, other termination blocks)
and bullets (used for wire to wire connection)